

Unit III

3

Regression

3.1 : Introduction of Regression

Q.1 Define and explain regression with its model.

[SPPU : Dec-17, End Sem, Marks 4]

Ans. : • Regression finds correlations between dependent and independent variables. If the desired output consists of one or more continuous variable, then the task is called as regression.

- Therefore, regression algorithms help predict continuous variables such as house prices, market trends, weather patterns, oil and gas prices etc.
- Fig. Q.1.1 shows regression.

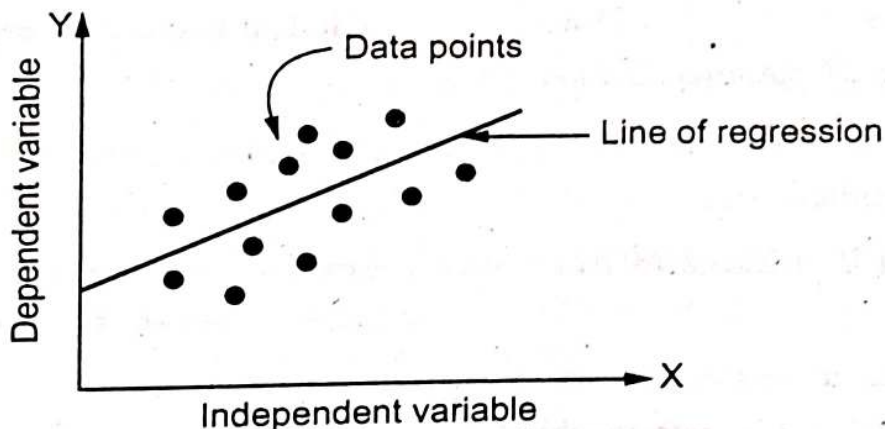


Fig. Q.1.1 Regression

- When the targets in a dataset are real numbers, the machine learning task is known as regression and each sample in the dataset has a real-valued output or target.
- Regression analysis is a set of statistical methods used for the estimation of relationships between a dependent variable and one or more independent variables. It can be utilized to assess the strength of

the relationship between variables and for modelling the future relationship between them.

- The two basic types of regression are linear regression and multiple linear regression.

3.2 : Univariate Regression

Q.2 Explain univariate regression.

Ans. : • Univariate data is the type of data in which the result depends only on one variable. If there is only one input variable then we call it 'Single Variable Linear Regression' or 'Univariate Linear Regression.'

- The function that we are trying to develop looks like this :

$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

$$y = mx + b$$

- That is because linear regression is essentially the algorithm for finding the line of best fit for a set of data.
- The algorithm finds the values for θ_0 and θ_1 that best fit the inputs and outputs given to the algorithm. This is called univariate linear regression because the ?? parameters only go up to 1.
- The univariate linear regression algorithm is much simpler than the one for multivariate.

Q.3 When is it suitable to use linear regression over classification ?

☞ [SPPU : Dec-16, End Sem, Marks 5]

Ans. : • Linear regression is a statistical method that allows us to summarize and study relationships between two continuous (quantitative) variables.

- The objective of a linear regression model is to find a relationship between the input variables and a target variable.
 1. One variable, denoted x , is regarded as the predictor, explanatory, or independent variable.
 2. The other variable, denoted y , is regarded as the response, outcome, or dependent variable.

- Regression models predict a continuous variable, such as the sales made on a day or predict temperature of a city. Let's imagine that we fit a line with the training points that we have. If we want to add another data point, but to fit it, we need to change existing model.
- This will happen with each data point that we add to the model; hence, linear regression isn't good for classification models.
- Regression estimates are used to explain the relationship between one dependent variable and one or more independent variables. Classification predicts categorical labels (classes), prediction models continuous - valued functions. Classification is considered to be supervised learning.
- Classifies data based on the training set and the values in a classifying attribute and uses it in classifying new data. Prediction means models continuous - valued functions, i.e. predicts unknown or missing values.

Q.4 Why do we need to regularize in regression? Explain.

[SPPU : Dec.-16, End Sem, Marks 5]

Ans. : • Regression model fails to generalize on unseen data. This could happen when the model tries to accommodate all kinds of changes in the data including those belonging to both the actual pattern and also the noise.

- As a result, the model ends up becoming a complex model having significantly high variance due to overfitting, thereby impacting the model performance (accuracy, precision, recall, etc.) on unseen data.

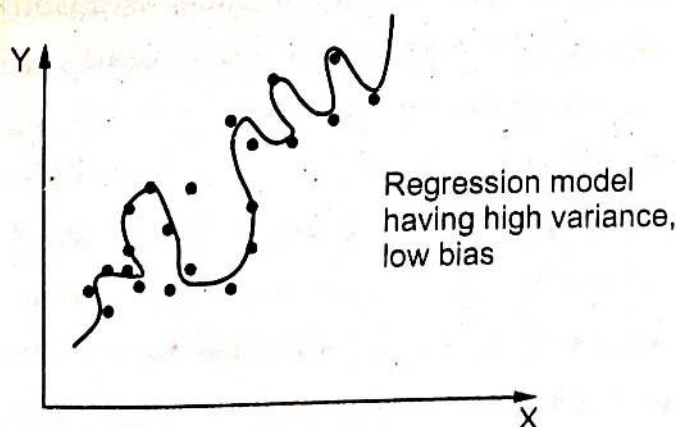


Fig. Q.4.1 Regression model before regularization

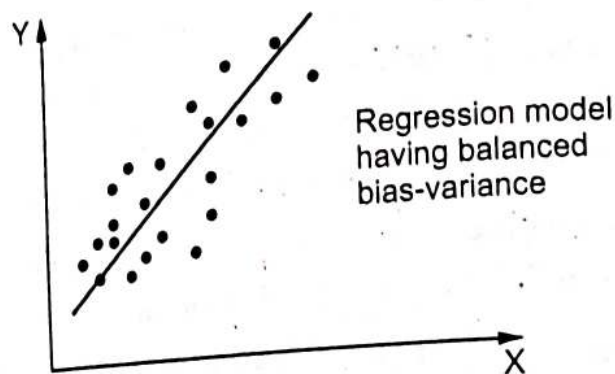


Fig. Q.4.2 Regression model after regularization

- Regularization needed for reducing overfitting in the regression model.
- Regularization techniques are used to calibrate the coefficients of the determination of multi-linear regression models in order to minimize the adjusted loss function.
- Regularization methods provide a means to control our regression coefficients, which can reduce the variance and decrease our of sample error.
- The goal is to reduce the variance while making sure that the model does not become biased (underfitting). After applying the regularization technique, the following model could be obtained.

Q.5 Explain least square method. [SPPU : May-16, End Sem, Marks 5]

OR What do you mean by least square method? Explain least square method in the context of linear regression.

[SPPU : Dec-19, End Sem, Marks 5]

Ans. : • The method of least squares is about estimating parameters by minimizing the squared discrepancies between observed data, on the one hand, and their expected values on the other.

- Considering an arbitrary straight line, $y = b_0 + b_1x$, is to be fitted through these data points. The question is "Which line is the most representative" ?
- What are the values of b_0 and b_1 such that the resulting line "best" fits the data points ? But, what goodness-of-fit criterion to use to determine among all possible combinations of b_0 and b_1 ?

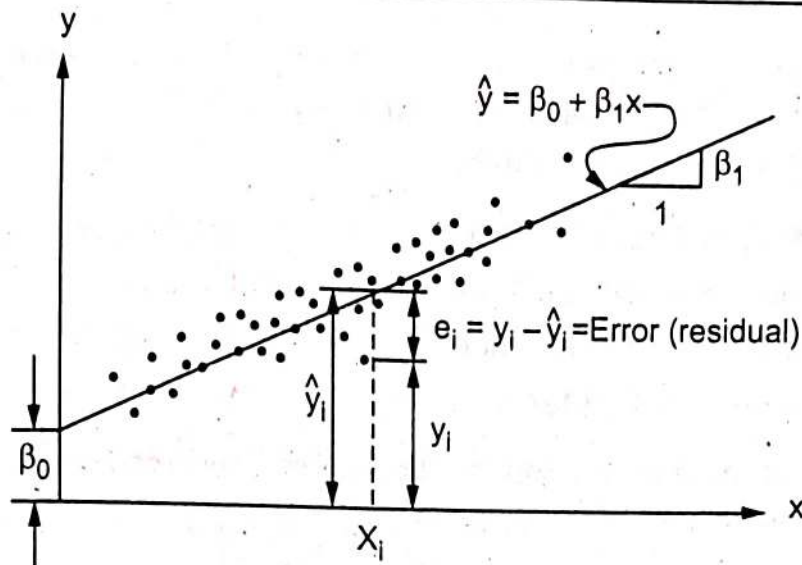


Fig. Q.5.1

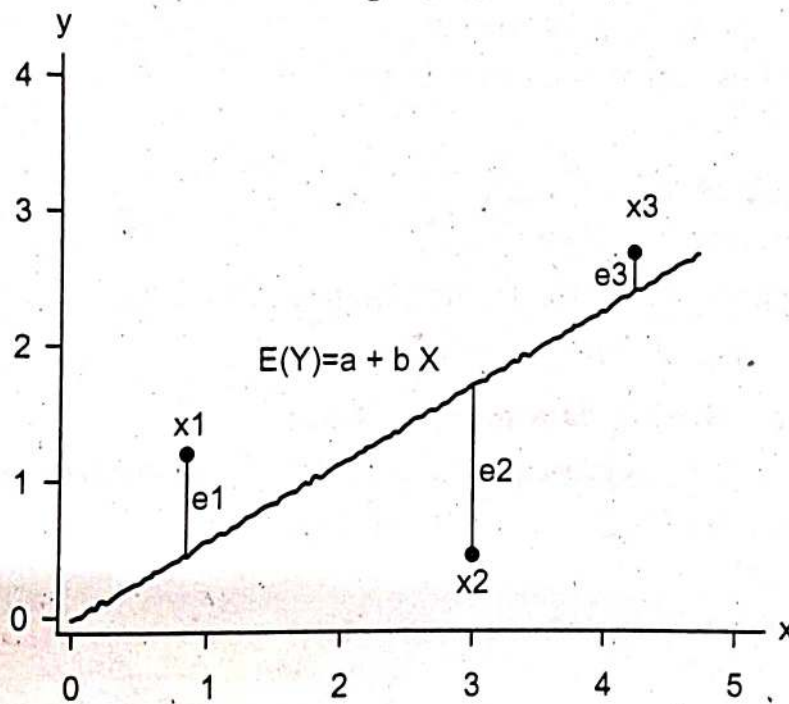


Fig. Q.5.2

- The Least Squares (LS) criterion states that the sum of the squares of errors is minimum. The least-squares solutions yields $y(x)$ whose elements sum to 1, but do not ensure the outputs to be in the range $[0,1]$.
- How to draw such a line based on data points observed ? Suppose a imaginary line of $y = a + bx$.
- Imagine a vertical distance between the line and a data point
 $E = Y - E(Y)$.

- This error is the deviation of the data point from the imaginary line, regression line. Then what is the best values of a and b ? A and b that minimizes the sum of such errors.
- Deviation does not have good properties for computation. Then why do we use squares of deviation? Let us get a and b that can minimize the sum of squared deviations rather than the sum of deviations. This method is called **least squares**.
- Least squares method minimizes the sum of squares of errors. Such a and b are called least squares estimators i.e. estimators of parameters α and β .
- The process of getting parameter estimators (e.g., a and b) is called **estimation**. Least squares method is the estimation method of Ordinary Least Squares (OLS).

Disadvantages of least square

1. Lack robustness to outliers
2. Certain datasets unsuitable for least squares classification
3. Decision boundary corresponds to ML solution

Q.6 Consider following data for 5 students.

Each X_i ($i = 1$ to 5) represents the score of i^{th} student in standard X and corresponding Y_i ($i = 1$ to 5) represents the score of i^{th} student in standard XII.

- i) What linear regression equation best predicts standard XIIth score?
- ii) Find regression line that fits best for given sample data.
- iii) How to interpret regression equation?
- iv) If a student's score is 80 in std X, then what is his expected score in XII standard?

Student	Score in X standard (X_i)	Score in XII standard (Y_i)
1	95	85
2	85	95

3	80	70
4	70	65
5	60	70

- Ans. : • The mean of the x values denoted by $\bar{X}$
- The mean of the y values denoted by $\bar{Y}$
 - The standard deviation of the x values (denoted S_X)
 - The standard deviation of the y values (denoted S_Y)

X_i	Y_i	X^2	XY	Y^2	$(X_i - \bar{X})$	$(Y_i - \bar{Y})$
95	85	9025	8075	7225	17	8
85	95	7225	8075	9025	7	18
80	70	6400	5600	4900	2	-7
70	65	4900	4550	4225	-8	-12
60	70	3600	4200	4900	-18	-7
$\Sigma X = 390$	$\Sigma Y = 385$	$\Sigma X^2 = 31150$	$\Sigma XY = 30500$	$\Sigma Y^2 = 30275$		

Find : $\bar{X}, \bar{Y}, S_X, S_Y$

$$\bar{X} = \frac{\sum X_i}{n}, \quad \bar{Y} = \frac{\sum Y_i}{n}, \quad S_X = \sqrt{\frac{\sum (X_i - \bar{X})^2}{n-1}}, \quad S_Y = \sqrt{\frac{\sum (Y_i - \bar{Y})^2}{n-1}}$$

$$\bar{X} = 390/5 = 78, \quad \bar{Y} = 385/5 = 77,$$

$$S_X = \sqrt{(390 - 78)^2 / 4} = \sqrt{24336} = 156$$

$$S_Y = \sqrt{(385 - 77)^2 / 4} = \sqrt{23716} = 154$$

We also need to compute the squares of the deviation scores :

X_i	Y_i	$(X_i - \bar{X})^2$	$(Y_i - \bar{Y})^2$	$(X_i - \bar{X}) \times (Y_i - \bar{Y})$
95	85	289	64	136
85	95	49	324	136
80	70	4	49	-14
70	65	64	144	96
60	70	324	49	126
$\Sigma X = 390$	$\Sigma Y = 385$	730	630	$\Sigma (X_i - \bar{X}) \times (Y_i - \bar{Y}) = 480$

The regression equation is a linear equation of the form : $\hat{Y} = \beta_0 + \beta_1 X$

First, we solve for the regression coefficient (β_1) :

$$\beta_1 = \frac{\sum [(X_i - \bar{X})(Y_i - \bar{Y})]}{\sum [(X_i - \bar{X})^2]}$$

$$= 480 / 730 = 0.657$$

Once we know the value of the regression coefficient (β_1), we can solve for the regression slope (β_0) :

$$\beta_0 = \bar{Y} - \beta_1 \bar{X}$$

$$\beta_0 = 385 - 0.657 * 390 = 128.77$$

Therefore, the regression equation is $\hat{Y} = 128.77 + 0.657 X$

Q.7 Consider following data :

i) Find values of β_0 and β_1 w.r.t. linear regression model which best fits given data.

ii) Interpret and explain equation of regression line.

iii) If new person rates "Bahubali -Part I" as 3 then predict the rating of same person for "Bahubali -Part II".

Person	X_i = Rating for movie "Bahubali Part - I" by i^{th} person	Y_i = Rating for movie "Bahubali Part - II" by i^{th} person
1 st	4	3
2 nd	2	4
3 rd	3	2
4 th	5	5
5 th	1	3
6 th	3	1

[SPPU : Aug.-18, In Sem, Marks 6]

Ans. :

X_i	Y_i	X^2	XY	Y^2	$(X_i - \bar{X})$	$(Y_i - \bar{Y})$
4	3	16	12	9	1	0
2	4	4	8	16	-1	1
3	2	9	6	4	0	-1
5	5	25	25	25	2	2
1	3	1	3	9	-2	0
3	1	9	3	1	0	-2
$\Sigma X = 18$	$\Sigma Y = 18$	$\Sigma X^2 = 64$	$\Sigma XY = 57$	$\Sigma Y^2 = 64$		

Find : $\bar{X}$, $\bar{Y}$, S_X , S_Y

$$\bar{X} = \frac{\sum X_i}{n}, \bar{Y} = \frac{\sum Y_i}{n}, S_X = \sqrt{\frac{\sum (X_i - \bar{X})^2}{n-1}}, S_Y = \sqrt{\frac{\sum (Y_i - \bar{Y})^2}{n-1}}$$

$$\bar{X} = 18/6 = 3, \bar{Y} = 18/6 = 3,$$

$$S_X = \sqrt{(18-3)^2 / 5} = \sqrt{45}$$

$$S_Y = \sqrt{(18-3)^2 / 5} = \sqrt{45}$$

We also need to compute the squares of the deviation scores :

X_i	Y_i	$(X_i - \bar{X})^2$	$(Y_i - \bar{Y})^2$	$(X_i - \bar{X}) \times (Y_i - \bar{Y})$
4	3	1	0	0
2	4	1	1	-1
3	2	0	1	0
5	5	4	4	4
1	3	4	0	0
3	1	0	4	0
$\Sigma X = 18$	$\Sigma Y = 18$	$\Sigma (X_i - \bar{X})^2 = 10$	$\Sigma (Y_i - \bar{Y})^2 = 10$	$\Sigma (X_i - \bar{X})(Y_i - \bar{Y}) = 3$

The regression equation is a linear equation of the form: $\hat{Y} = \beta_0 + \beta_1 X$

First, we solve for the regression coefficient (β_1):

$$\beta_1 = \frac{\sum [(X_i - \bar{X})(Y_i - \bar{Y})]}{\sum [(X_i - \bar{X})^2]}$$

$$= 3/10 = 0.3$$

Once we know the value of the regression coefficient (β_1), we can solve for the regression slope (β_0):

$$\beta_0 = \bar{Y} - \beta_1 \bar{X}$$

$$\beta_0 = 18 + 0.3 * 18 = 12.6$$

Therefore, the regression equation is $\hat{y} = 12.6 + 0.3 X$

3.3 : Cost Function

Q.8 Define and explain Squared Error (SE) and Mean Squared Error (MSE) w.r.t Regression. [SPPU : May-18, End Sem, Marks 5]

- Ans. : • The most common measurement of overall error is the sum of the squares of the errors, or SSE (sum of squared errors). The line with the smallest SSE is called the least-squares regression line.
- Mean Squared Error (MSE) is calculated by taking the average of the square of the difference between the original and predicted values of the data. It can also be called the quadratic cost function or sum of squared errors.
 - The value of MSE is always positive or greater than zero. A value close to zero will represent better quality of the estimator/predictor. An MSE of zero (0) represents the fact that the predictor is a perfect predictor.

$$MSE = \frac{1}{N} \sum_{i=1}^N (\text{Actual values} - \text{Predicted values})^2$$

- Here N is the total number of observations/rows in the dataset. The sigma symbol denotes that the difference between actual and predicted values taken on every i value ranging from 1 to n.
- Mean squared error is the most commonly used loss function for regression. MSE is sensitive towards outliers and given several examples with the same input feature values, the optimal prediction will be their mean target value. This should be compared with Mean

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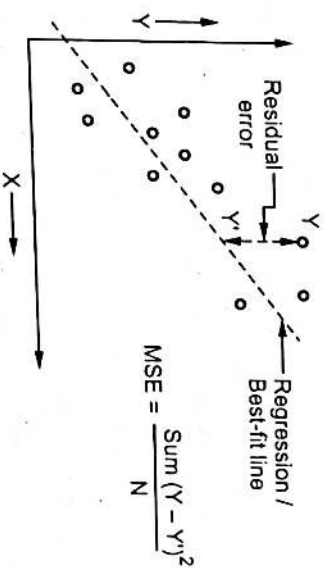


Fig. Q.8.1 Representation of MSE

Absolute Error, where the optimal prediction is the median. MSE is thus good to use if you believe that your target data, conditioned on the input, is normally distributed around a mean value, and when it's important to penalize outliers extra much.

- MSE incorporates both the variance and the bias of the predictor. MSE also gives more weight to larger differences. The bigger the error, the more it is penalized.

- Example : You want to predict future house prices. The price is a continuous value, and therefore we want to do regression. MSE can here be used as the loss function

Q.9 How the performance of a regression function is measured ?

[SPPU : Dec-17, End Sem, Marks 4]

Ans. : • Following are the performance metrics used for evaluating a regression model :

- a) Mean Absolute Error (MAE)
- b) Mean Squared Error (MSE)
- c) Root Mean Squared Error (RMSE)
- d) R-squared
- e) Adjusted R-squared

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1. MAE :

- MAE is the sum of absolute differences between our target and predicted variables. So it measures the average magnitude of errors in a set of predictions, without considering their directions.

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

2. Mean Squared Error (MSE)

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

3. Root Mean Square Error (RMSE)

- Root Mean Square Error (RMSE) is a standard way to measure the error of a model in predicting quantitative data.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

4. R-squared

- R-squared is also known as the coefficient of determination. This metric gives an indication of how good a model fits a given dataset. It indicates how close the regression line is to the actual data values.
- The R squared value lies between 0 and 1 where 0 indicates that this model doesn't fit the given data and 1 indicates that the model fits perfectly to the dataset provided.

$$R\text{-squared} = 1 - \frac{\text{First sum of errors}}{\text{Second sum of errors}}$$

$$R^2 = 1 - \frac{SS_{RES}}{SS_{TOT}} = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}$$

5) Adjusted R-squared :

- The adjusted R-squared shows whether adding additional predictors improve a regression model or not.

$$\text{Adjusted } R^2 = 1 - \frac{(1 - R^2)(N - 1)}{N - p - 1}$$

Q.10 For a given data having 100 examples, if squared errors SE_1 , SE_2 , and SE_3 are 13.33, 3.33 and 4.00 respectively, calculate Mean Squared Error (MSE). State the formula for MSE.

Ans. : [SPPU : May-16, End Sem, Marks 5]

Ans. :

$$\text{Squared Error}_1 + \text{Squared Error}_2$$

$$\text{Mean Squared Error} = \frac{+ \dots + \text{Squared Error}_N}{\text{Number of data samples}}$$

$$\text{Mean Squared Error} = \frac{13.33 + 3.33 + 4.00}{100} = 0.2066$$

Q.11 What do you mean by coefficient of regression? Explain SST, SSE, SSR, MSE in the context of regression.

Ans. : [SPPU : Dec-19, End Sem, Marks 5]

Ans. : • The quantities multiplied by the variables in a regression equation are called regression coefficients. Linear regression is the most common type of regression. The goal of linear regression is to determine which regression coefficients provide the best-fitting line.

• In linear regression, the regression coefficients assist in estimating the value of an unknown variable using a known variable. The regression coefficients analyses how the variables are dependent on other.

• Regression line of X on Y gives the best estimate for the value of X for any specific given values of Y :

$$X = a + b Y$$

where a = X - intercept

b = Slope of the line

X = Dependent variable

Y = Independent variable

• Also Refer Q.8 and Q.9.

3.4 : Performance Evaluation

Q.12 Write short note on clustering.

Ans. : • Clustering groups data points based on their similarities. Each group is called a cluster and contains data points with high similarity and low similarity with data points in other clusters.

• The objective of clustering is to segregate groups with similar traits and bundle them together into different clusters.

• Silhouette analysis can be used to study the separation distance between the resulting clusters. The silhouette plot displays a measure of how close each point in one cluster is to points in the neighboring clusters. This measure has a range of $[-1, 1]$.

• Silhouette coefficients near + 1 indicate that the sample is far away from the neighboring clusters. A value of 0 indicates that the sample is on or very close to the decision boundary between two neighboring clusters and negative values indicate that those samples might have been assigned to the wrong cluster.

• Many clustering algorithms use distance measures to determine the similarity or dissimilarity between any pair of data points. A valid distance measure should be symmetric and obtains its minimum value (usually zero) in case of identical data points. By computing the distance or (dis) similarity between each pair of observations, a dissimilarity or distance matrix is obtained.

3.5 : Optimizing Simple Linear Regression with Gradient Descent Algorithm

Q.13 List the derivative based optimization methods.

Ans. : • Derivative based optimization methods are used for :

1. Optimization of nonlinear neuro-fuzzy models
2. Neural network learning
3. Regression analysis in nonlinear models

Q.14 Explain gradient descent algorithm. Also explain its limitation.

Ans. : • Gradient descent is a first-order optimization algorithm. To find a local minimum of a function using gradient descent, one takes steps proportional to the negative of the gradient of the function at the current point.

• Gradient descent is popular for very large-scale optimization problems because it is easy to implement, can handle black box functions, and each iteration is cheap.

• Given a differentiable scalar field $f(x)$ and an initial guess x_1 , gradient descent iteratively moves the guess toward lower values of "f" by taking steps in the direction of the negative gradient $-\nabla f(x)$.

• Locally, the negated gradient is the steepest descent direction, i.e., the direction that x would need to move in order to decrease "f" the fastest. The algorithm typically converges to a local minimum, but may rarely reach a saddle point, or not move at all if x_1 lies at a local maximum.

• The gradient will give the slope of the curve at that x and its direction will point to an increase in the function. So we change x in the opposite direction to lower the function value :

$$X_{k+1} = x_k - \lambda \nabla f(x_k)$$

• The $\lambda > 0$ is a small number that forces the algorithm to make small jumps

Limitations of gradient descent :

• Gradient descent is relatively slow close to the minimum : technically, its asymptotic rate of convergence is inferior to many other methods.

• For poorly conditioned convex problems, gradient descent increasingly 'zigzags' as the gradients point nearly orthogonally to the shortest direction to a minimum point

Q.15 Explain steepest descent method.

Ans. : • Steepest descent is also known as gradient method.

- This method is based on first order Taylor series approximation of objective function. This method is also called saddle point method. Fig. Q.15.1 shows steepest descent method.

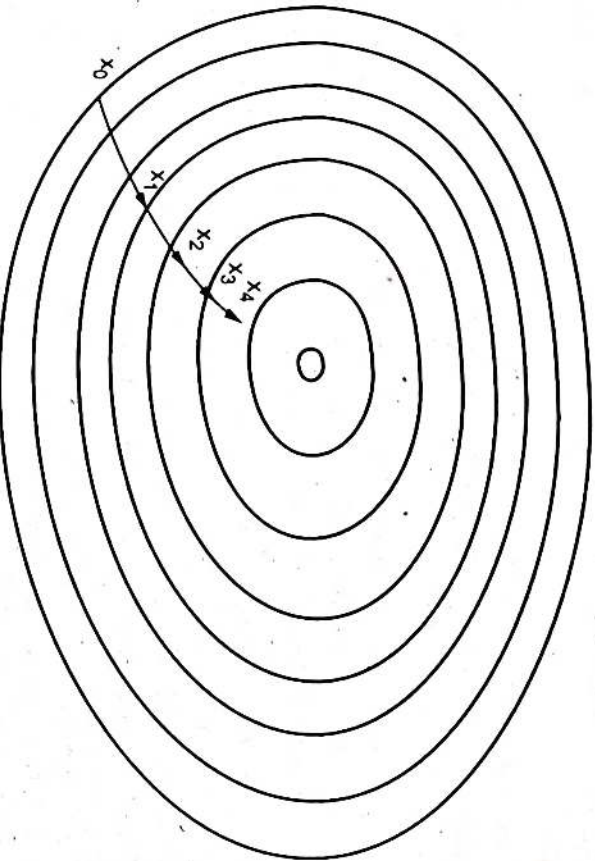


Fig. Q.15.1 Steepest descent method

- The steepest descent is the simplest of the gradient methods. The choice of direction is where f decreases most quickly, which is in the direction opposite to $\nabla f(x)$. The search starts at an arbitrary point x_0 and then go down the gradient, until reach close to the solution.
- The method of steepest descent is the discrete analogue of gradient descent, but the best move is computed using a local minimization rather than computing a gradient. It is typically able to converge in few steps but it is unable to escape local minima or plateaus in the objective function.
- The gradient is everywhere perpendicular to the contour lines. After each line minimization the new gradient is always orthogonal to the previous step direction. Consequently, the iterates tend to zig-zag down the valley in a very inefficient manner.

- The method of steepest descent is simple, easy to apply and each iteration is fast. It also very stable; if the minimum points exist, the method is guaranteed to locate them after at least an infinite number of iterations.

3.6 : Multivariate Regression

Q.16 What is multivariate regression ?

[SPPU : Oct-16, In Sem, Marks 5]

- Ans. : • Multivariate regression involves multiple data variables for analysis. It is a supervised machine learning algorithm.
- Multivariate regression is used when there is a need to predict the value of a variable which is based on the value of two more variables. The value of the variable which is needed to be predicted is called the dependent variable.
 - If multiple independent variables affect the response variable, then the analysis calls for a model different from that used for the single predictor variable. In a situation where more than one independent factor (variable) affects the outcome of a process, a multiple regression model is used. This is referred to as multiple linear regression model or multivariate least squares fitting.

- Let $z_1, z_2, \dots, z_r$ be a set of r predictors believed to be related to a response variable Y . The linear regression model for the j^{th} sample unit has the form

$$Y_j = \beta_0 + \beta_1 z_{j1} + \beta_2 z_{j2} + \dots + \beta_r z_{jr} + \epsilon_j$$

where ϵ is a random error and $\beta_i, i = 0, 1, \dots, r$ are unknown regression coefficients.

- With n independent observations, we can write one model for each sample unit so that the model is now

$$Y = Z\beta + \epsilon$$

where Y is $n \times 1$, Z is $n \times (r+1)$, β is $(r+1) \times 1$ and ϵ is $n \times 1$

- In order to estimate β , we take a least squares approach that is analogous to what we did in the simple linear regression case.
- In matrix form, we can arrange the data in the following form :

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1k} \\ 1 & x_{21} & x_{22} & \dots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{N1} & x_{N2} & \dots & x_{Nk} \end{bmatrix}, Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}, \hat{a} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}$$

- where β_j are the estimates of the regression coefficients.

Q.17 What is multivariate regression? How will it be different from Univariate regression ?

☞ [SPPU : May-16, End Sem, Marks 5, Oct.-18, In Sem, Marks 4]

Ans. :

Simple regression	Multiple regression
One dependent variable Y predicted from one independent variable X	One dependent variable Y predicted from a set of independent variables ($X_1, X_2 \dots X_k$)
One regression coefficient	One regression coefficient for each independent variable
r^2 : Proportion of variation in dependent variable Y predictable from X	R^2 : Proportion of variation in dependent variable Y predictable by set of independent variables (X 's)

- Also Refer Q.16.

Q.18 What do you mean by zero-centering ?

☞ [SPPU, OCT.-16, In Sem, Marks 5]

OR What do you mean by zero centered and un-correlated features ? What is the use of it in the solution of multivariate linear regression ?

☞ [SPPU : May-18, End Sem, Marks 6]

Ans. : • Feature normalization is often required to neutralize the effect of different quantitative features being measured on different scales. If the

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features are approximately normally distributed, we can convert them into z-scores by centring on the mean and dividing by the standard deviation. If we don't want to assume normality we can centre on the median and divide by the interquartile range.

- Sometimes feature normalization is understood in the stricter sense of expressing the feature on a [0,1] scale. If we know the feature's highest and lowest values h and l , then we can simply apply the linear scaling.
- Feature calibration is understood as a supervised feature transformation adding a meaningful scale carrying class information to arbitrary features. This has a number of important advantages. For instance, it allows models that require scale, such as linear classifiers, to handle categorical and ordinal features. It also allows the learning algorithm to choose whether to treat a feature as categorical, ordinal or quantitative.

- The goal of both types of normalization is to make it easier for your learning algorithm to learn. In feature normalization, there are two standard things to do :
1. Centering : Moving the entire data set so that it is centered around the origin.
2. Scaling : Rescaling each feature so that one of the following holds :
a) Each feature has variance 1 across the training data.
b) Each feature has maximum absolute value 1 across the training data.
- The goal of centering is to make sure that no features are arbitrarily large.

Q.19 List the characteristics of multivariate regression.

Ans. : Characteristics of multivariate regression are as follows :

- Multivariate regression allows one to have a different view of the relationship between various variables from all the possible angles.
- It helps you to predict the behavior of the response variables depending on how the predictor variables move.

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- Multivariate regression can be applied to various machine learning fields, economic, science and medical research studies.

Q.20 What is difference between regression and correlation ?

Ans. :

Regression	Correlation
Regression tells us how to draw the straight line described by the correlation	Correlation describes the strength of a linear relationship between two variables
For regression only the dependent variable Y must be random.	For correlation, both variables should be random variables
Main goal is use the measure of relation to predict values of the random variable based on values of the fixed variable.	Main goal is simply to find a number that expresses the relation between the variables

3.7 : Introduction to Polynomial Regression

Q.21 What is a polynomial regression? How it can be represented in a form of a matrix.

Ans. : • A polynomial regression consists of constants and variables that are combined using operations such as addition, subtraction and multiplication. Polynomial regression is a technique based on a trick that allows the use of linear models even when the dataset has strong non-linearities.

- With polynomial regression, the data is approximated using a polynomial function. A polynomial is a function that takes the form $f(x) = c_0 + c_1 x + c_2 x^2 \dots c_n x^n$ where n is the degree of the polynomial and c is a set of coefficients.

- A polynomial of degree 0 (zero) is just a constant because $f(x) = c_0 x_0 = c_0$. Likewise performing polynomial regression with a degree of 0 (zero) on a set of data returns a single constant value. It is the same as the mean average of that data.

- Linear regression is polynomial regression of degree 1, and generally takes the form $y = m x + b$ where m is the slope, and b is the y-intercept. It could just as easily be written $f(x) = c_0 + c_1 x$ with c_1 being the slope and c_0 the y-intercept.

- The order of the polynomial regression model depends on the number of features included in the model, so a model with m features is an n^{th} - degree or n^{th} - order polynomial regression. The general form of the design matrix with m -degrees looks like this :

$$\begin{bmatrix} 1 & x_1 & x_1^2 & \dots & x_1^m \\ 1 & x_2 & x_2^2 & \dots & x_2^m \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^m \end{bmatrix}$$

- The matrix is always invertible as they follow the statistical rule of $m < n$ and thus become Vandermonde matrix. While it might be tempting to fit the curve and decrease error, it is often required to analyze whether fitting all the points makes sense logically and avoid overfitting.
- This is a highly important step as Polynomial Regression despite all its benefit is still only a statistical tool and requires human logic and intelligence to decide on right and wrong.

Q.22 Explain overfitting and underfitting.

Ans. : **Overfitting**

- Training error can be reduced by making the hypothesis more sensitive to training data, but this may lead to overfitting and poor generalization.
- Overfitting occurs when a statistical model describes random error or noise instead of the underlying relationship. Low error rates and a high variance are good indicators of overfitting. Fig. Q.22.1 shows overfitting.

- Overfitting is when a classifier fits the training data too tightly. Such a classifier works well on the training data but not on independent test data. It is a general problem that plagues all machine learning methods.

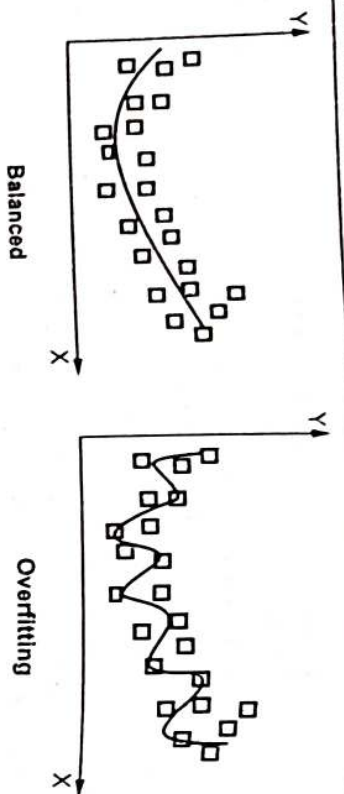


Fig. Q.22.1

- Overfitting generally occurs when a model is excessively complex, such as having too many parameters relative to the number of observations.
- We can determine whether a predictive model is under-fitting or overfitting the training data by looking at the prediction error on the training data and the evaluation data.

Reasons for overfitting

1. Noisy data
2. Training set is too small
3. Large number of features

Underfitting :

- Underfitting happens when the learner has not found a solution that fits the observed data to an acceptable level.

- Underfitting : If we put too few variables in the model, leaving out variables that could help explain the response, we are underfitting. Consequences :

1. Fitted model is not good for prediction of new data -prediction is biased
2. Regression coefficients are biased

Underfitting examples :

1. The learning time may be prohibitively large, and the learning stage was prematurely terminated.
2. The learner did not use a sufficient number of iterations.

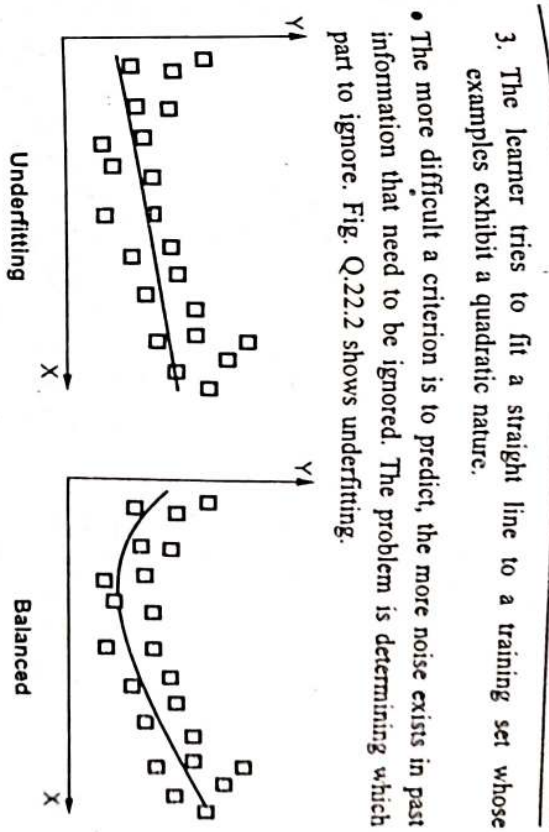


Fig. Q.22.2

How do we know if we are underfitting or overfitting ?

1. If by increasing capacity we decrease generalization error, then we are underfitting, otherwise we are overfitting.
2. If the error in representing the training set is relatively large and the generalization error is large, then underfitting.
3. If the error in representing the training set is relatively small and the generalization error is large, then overfitting.
4. There are many features but relatively small training set.

Q.23 Explain bias-variance trade off.

Ans. : • In the experimental practice we observe an important phenomenon called the bias variance dilemma.

- In supervised learning, the class value assigned by the learning model built based on the training data may differ from the actual class value. This error in learning can be of two types, errors due to 'bias' and error due to 'variance'.

- Fig. Q.23.1 shows bias-variance trade off.

- Give two classes of hypothesis (e.g. linear models and k-NNs) to fit to some training data set, we observe that the more flexible hypothesis

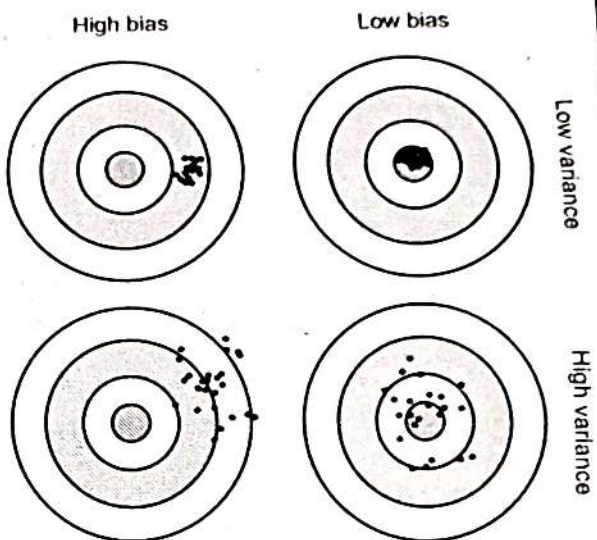


Fig. Q.23.1 Bias-variance trade off

class has a low bias term but a higher variance term. If we have parametric family of hypothesis, then we can increase the flexibility of the hypothesis but we still observe the increase of variance.

- The bias-variance-dilemma is the problem of simultaneously minimizing two sources of error that prevent supervised learning algorithm from generalizing beyond their training set :

 1. The bias is error from erroneous assumptions in the learning algorithm. High bias can cause an algorithm to miss the relevant relations between features and target outputs.
 2. The variance is error from sensitivity to small fluctuations in the training set. High variance can cause overfitting : modeling the random noise in the training data, rather than the intended outputs.

 - In order to reduce the model error, the designer can aim at reducing either the bias or the variance, as the noise components is irreducible.
 - As the model increases in complexity, its bias is likely to diminish. However, as the number of training examples is kept fixed, the

parametric identification of the model may strongly vary from one DN to another. This will increase the variance term.

- At one stage, the decrease in bias will be inferior to the increase in variance, warning that the model should not be too complex. Conversely, to decrease the variance term, the designer has to simplify its model so that it is less sensitive to a specific training set. This simplification will lead to a higher bias.

Q.24 Explain the above Fig. Q.24.1 (a), (b) and (c).

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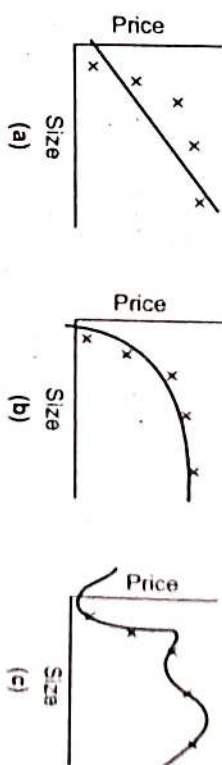


Fig. Q.24.1

Ans. : • Given Fig. Q.24.1 is related to overfitting and underfitting.

Underfitting (High bias and low variance) :

- A statistical model or a machine learning algorithm is said to have underfitting when it cannot capture the underlying trend of the data.
- It usually happens when we have less data to build an accurate model and also when we try to build a linear model with a non-linear data.

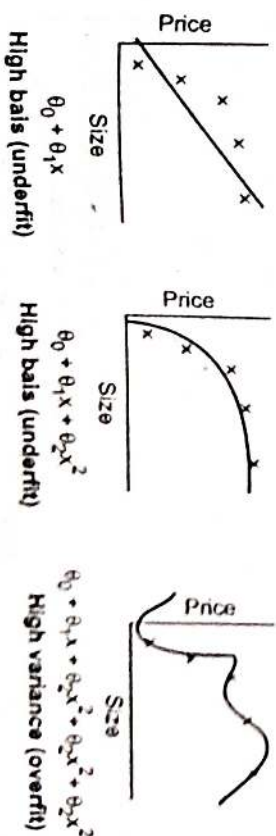


Fig. Q.24.2

- In such cases the rules of the machine learning model are too easy and flexible to be applied on such minimal data and therefore the model will probably make a lot of wrong predictions.

- Underfitting can be avoided by using more data and also reducing the features by feature selection.

Overfitting (High variance and low bias) :

- A statistical model is said to be overfitted, when we train it with a lot of data.
- When a model gets trained with so much of data, it starts learning from the noise and inaccurate data entries in our data set.
- Then the model does not categorize the data correctly, because of too many details and noise.
- The causes of overfitting are the non-parametric and non-linear methods because these types of machine learning algorithms have more freedom in building the model based on the dataset and therefore they can really build unrealistic models.
- A solution to avoid overfitting is using a linear algorithm if we have linear data or using the parameters like the maximal depth if we are using decision trees.

Q.25 Explain difference between clustering and classification.

Ans. :

Clustering	Classification
This function maps the data into one of several clusters which is the grouping of data items based on the similarities between them.	This model function classifies the data into one of several predefined categorical classes.
Involved in unsupervised learning	Involved in supervised learning
Training sample is not provided	Training sample is provided

The number of cluster is not known before clustering. These are identified after the completion of clustering.

The number of classes is known before classification as there is predefined output based on input data.

Data is not labeled.

Labeled data points.

Asks how can I group this set of items?

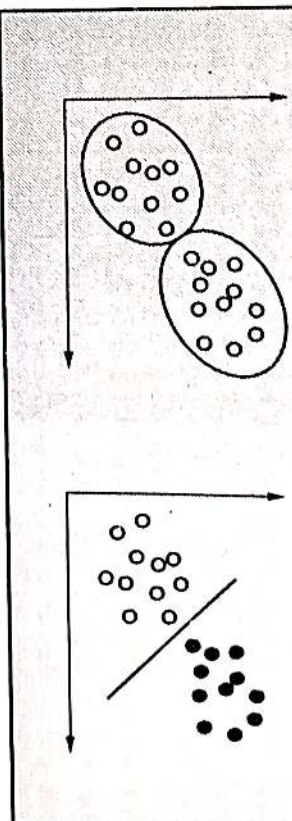
Asks what class does this item belong to?

Unknown number of classes

Known number of classes

Used to understand data

Used to classify future observations



END... ✍

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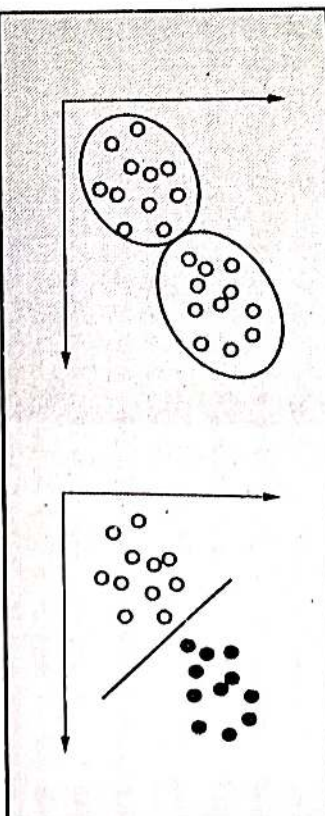
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END... 25